```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import accuracy_score, mean_squared_error
```

```
df=pd.read_csv('New york city fare prediction.csv')
```

```
df.head()
```

	key	fare_amount	pickup_datetime	pickup_longitude	pickup_latitude	dropoff_longitude	dropoff_latitude
0	2009-06-15 17:26:21.0000001	4.5	2009-06-15 17:26:21 UTC	-73.844311	40.721319	-73.841610	40.712278
1	2010-01-05 16:52:16.0000002	16.9	2010-01-05 16:52:16 UTC	-74.016048	40.711303	-73.979268	40.782004
2	2011-08-18 00:35:00.00000049	5.7	2011-08-18 00:35:00 UTC	-73.982738	40.761270	-73.991242	40.750562
3	2012-04-21 04:30:42.0000001	7.7	2012-04-21 04:30:42 UTC	-73.987130	40.733143	-73.991567	40.758092
4	2010-03-09 07:51:00.000000135	5.3	2010-03-09 07:51:00 UTC	-73.968095	40.768008	-73.956655	40.783762

Next steps: Generate code with df New interactive sheet

```
df.tail()
```

	key	fare_amount	pickup_datetime	pickup_longitude	pickup_latitude	dropoff_longitude	dropoff_latitude
49995	2013-06-12 23:25:15.0000004	15.0	2013-06-12 23:25:15 UTC	-73.999973	40.748531	-74.016899	40.70599
49996	2015-06-22 17:19:18.0000007	7.5	2015-06-22 17:19:18 UTC	-73.984756	40.768211	-73.987366	40.76059
49997	2011-01-30 04:53:00.00000063	6.9	2011-01-30 04:53:00 UTC	-74.002698	40.739428	-73.998108	40.75948
49998	2012-11-06 07:09:00.00000069	4.5	2012-11-06 07:09:00 UTC	-73.946062	40.777567	-73.953450	40.77968
49999	2010-01-13 08:13:14.0000007	10.9	2010-01-13 08:13:14 UTC	-73.932603	40.763805	-73.932603	40.76380

```
df.shape
```

```
(50000, 8)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 8 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   key                50000 non-null  object 
 1   fare_amount        50000 non-null  float64
 2   pickup_datetime    50000 non-null  object 
 3   pickup_longitude   50000 non-null  float64
 4   pickup_latitude    50000 non-null  float64
 5   dropoff_longitude  50000 non-null  float64
 6   dropoff_latitude   50000 non-null  float64
 7   passenger_count    50000 non-null  int64  
dtypes: float64(5), int64(1), object(2)
memory usage: 3.1+ MB
```

```
df.isnull().sum()
```

	0
key	0
fare_amount	0
pickup_datetime	0
pickup_longitude	0
pickup_latitude	0
dropoff_longitude	0
dropoff_latitude	0
passenger_count	0

dtype: int64

```
df
```

	key	fare_amount	pickup_datetime	pickup_longitude	pickup_latitude	dropoff_longitude	dropoff_latitud
0	2009-06-15 17:26:21.0000001	4.5	2009-06-15 17:26:21 UTC	-73.844311	40.721319	-73.841610	40.71227
1	2010-01-05 16:52:16.0000002	16.9	2010-01-05 16:52:16 UTC	-74.016048	40.711303	-73.979268	40.78200
2	2011-08-18 00:35:00.00000049	5.7	2011-08-18 00:35:00 UTC	-73.982738	40.761270	-73.991242	40.75056
3	2012-04-21 04:30:42.0000001	7.7	2012-04-21 04:30:42 UTC	-73.987130	40.733143	-73.991567	40.75809
4	2010-03-09 07:51:00.000000135	5.3	2010-03-09 07:51:00 UTC	-73.968095	40.768008	-73.956655	40.78376
...	...	...	...	...	...	...	.
49995	2013-06-12 23:25:15.0000004	15.0	2013-06-12 23:25:15 UTC	-73.999973	40.748531	-74.016899	40.70599
49996	2015-06-22 17:19:18.0000007	7.5	2015-06-22 17:19:18 UTC	-73.984756	40.768211	-73.987366	40.76059
49997	2011-01-30 04:53:00.00000063	6.9	2011-01-30 04:53:00 UTC	-74.002698	40.739428	-73.998108	40.75948
49998	2012-11-06 07:09:00.00000069	4.5	2012-11-06 07:09:00 UTC	-73.946062	40.777567	-73.953450	40.77968
49999	2010-01-13 08:13:14.0000007	10.9	2010-01-13 08:13:14 UTC	-73.932603	40.763805	-73.932603	40.76380

0000 rows × 8 columns

Next steps: Generate code with df New interactive sheet

```
df=pd.read_csv('New york city fare prediction.csv')
x=df.drop(columns=['fare_amount', 'key', 'pickup_datetime'],axis=1)
```

```
y=df['fare_amount']
```

```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
```

```
model=LinearRegression()
model.fit(x_train,y_train)
```

```
▾ LinearRegression
LinearRegression()
```

```
y_pred=model.predict(x_test)
mse=mean_squared_error(y_test,y_pred)
r2=model.score(x_test,y_test)
print('Mean Squared Error:',mse)
print('R-squared:',r2)
```

```
Mean Squared Error: 92.94908043572134
R-squared: 0.00023484981275878614
```

```
new = pd.DataFrame({
    'pickup_longitude': [-74.005973],
    'pickup_latitude': [40.712776],
    'dropoff_longitude': [-73.98642],
    'dropoff_latitude': [40.762121],
    'passenger_count': [1]
})
predicted_fare = model.predict(new)
print('Predicted fare:', predicted_fare)
```

```
Predicted fare: [11.28546241]
```